MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth Semester- Discipline Specific Core Course (DSC)

MG4DSCBTS200 – GENETIC ENGINEERING AND PLANT TISSUE CULTURE

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

ANSWER KEY

Part A – Fill in the blanks (Compulsory) (1 × 10 = 10 marks)

1. The enzyme used in PCR is _____ DNA polymerase.
   Answer: Taq DNA polymerase
2. Hybridization is based on complementary _____ pairing.
   Answer: base
3. _____ is the optimal pH of plant tissue culture media.
   Answer: 5.6–5.8 (or 5.7)
4. _____ proteins help plants from temperature stress.
   Answer: Heat shock
5. _____ is an application of ovary culture.
   Answer: Haploid production (also acceptable: seedless fruit development)
6. Terminal transferase adds nucleotides to the _____ end of DNA.
   Answer: 3′ end
7. Monnier developed a method for the development of _____.
   Answer: Haploid embryos / haploid plants
8. The first commercialized Bt crop was _____.
   Answer: Bt cotton
9. The repair mechanism that causes insertions or deletions is _____.
   Answer: Non-homologous end joining (NHEJ)
10. The plasmid vector pBR322 was constructed by _____ and Bolivar.
    Answer: Herbert Boyer

Part B – Short Answer (Answer any 10) (2 × 10 = 20 marks)

11. Mention two applications of cDNA libraries.
- **Write any two applications (1mark each)**
  Ex: Cloning of eukaryotic genes, Expression of proteins in bacteria

12. List any two applications of antisense therapy.
- **Write any two applications (1mark each)**
  Ex: Gene silencing in genetic diseases, Cancer therapy

13. What is the significance of somatic embryogenesis?
- **Rapid clonal multiplication**
- **Production of disease-free plants –(any 2 significance 1 mark each)**

14. Give two examples of somatic hybrid.
- **Write any two examples (1mark each)**
  ex: Pomato (potato × tomato), Citrus hybrids

15. What is lipofection?
- **A gene transfer method using lipid vesicles (liposomes) to deliver DNA into cells.**

16. Name the Indian scientists who pioneered the technique of triploid production.
- **Panchanan Maheshwari and S. S. Guha**

17. Mention any two methods used for quantification of DNA and RNA.
- **Write any two methods (1mark each)**
  Ex: UV spectrophotometry, Fluorometric assays

18. Explain electrofusion of protoplasts.
- **Fusion of plant protoplasts is induced by electric pulses that cause membrane breakdown and merging of cytoplasm.**

19. What is herbicide resistance in transgenic plants?
- **Introduction of genes that allow plants to survive herbicide application without damage.**

20. Describe one application of single-cell clones in biotechnology.
- **Production of genetically uniform plant populations through clonal propagation.**

21. Why is RNA more difficult to isolate than DNA?
- **RNA is unstable and easily degraded by ubiquitous RNases.**

22. Differentiate between kinase and phosphatase.

Kinase	Phosphatase
Adds phosphate group	Removes phosphate group
Uses ATP	Releases inorganic phosphate

Part C – Short Essay (Answer any 5) (4 × 5 = 20 marks)

23. Describe the steps involved in micropropagation.
- **Selection of explant, Surface sterilisation, Initiation of culture, Shoot multiplication, Rooting, Acclimatisation and transfer to soil**

24. Discuss the advantages and limitations of Sanger sequencing.
- **Advantages: High accuracy, Suitable for small DNA fragments (any 2)**
- **Limitations: Time-consuming, Expensive for large genomes, Low throughput compared to NGS (any 2)**

25. Explain the differences in detection methods used in Southern and Western blotting.
- **Any 2 differences**

Southern blot	Western blot
Detects DNA	Detects proteins
Uses DNA probes	Uses antibodies
Hybridization detection	Immunodetection

26. Discuss the scope of genetic engineering in agriculture.
- **Discuss any 4 scopes**

Ex: Development of pest-resistant crops, Herbicide-tolerant crops, Improved nutritional quality, Abiotic stress tolerance, Increased yield and shelf life

28. Explain the types of gene therapy.
- **Somatic gene therapy – genes introduced into body cells; not inherited**
- **Germline gene therapy – genes introduced into gametes or embryos; heritable**

28. Describe the method of cryopreservation.
- **Selection of plant material**
- **Treatment with cryoprotectants**
- **Rapid freezing in liquid nitrogen (−196 °C)**
- **Storage**
- **Thawing and regeneration**

29. What is callus culture? What are the applications of callus culture?
- **Callus culture: Growth of unorganized, undifferentiated plant cells on nutrient medium under sterile conditions.**
- **Applications: Plant regeneration, Genetic transformation studies, Production of secondary metabolites, Somaclonal variation for crop improvement (any 2)**